Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I can use what I learned about variables, conditionals, loops, and functions. I want to improve my skills using Drafter to create an interactive website. This way, I can produce more useful programs and use computer science to a greater extent.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest part will probably be asking the user for the data to set up the trajectory equation. The hardest part will be calculating the ball's trajectory accurately and figuring out whether it will go into the hoop.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

Calculating the ball's path and creating the graph will probably take the most time. I

should first build the basic website and input fields, then work on the physics calculations. After that, I can add the graph and shot prediction.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions like my phone and other assignments, so I will try to work in a quiet place and manage my time. I will need the library and a set time each week to work on this assignment. I do not have any more questions right now.

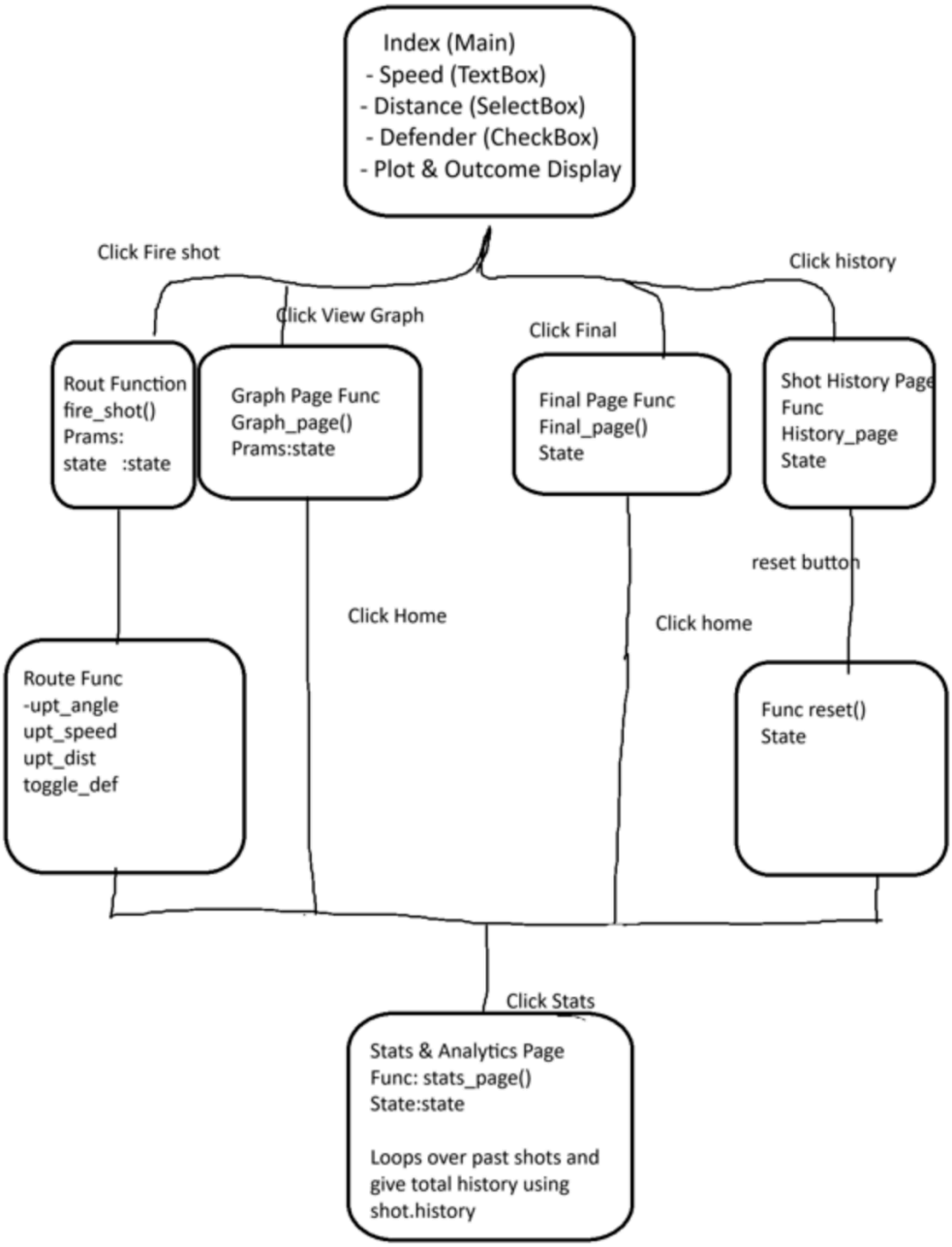
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Index (Main)
- Speed (TextBox)
- Distance (SelectBox)
- Defender (CheckBox)
- Plot & Outcome Display
Click Fire shot
Click View Graph
Click Final
Click history
Rout Function
fire_shot()
Prams:
state :state
Graph Page Func
Graph_page()
Prams:state
Final Page Func
Final_page()
State
Shot History Page
Func
History_page
State
reset button
Click Home
Click home
Route Func
-upt_angle
upt_speed
upt_dist
toggle_def
Func reset()
State
Click Stats
Stats & Analytics Page
Func: stats_page()
State:state
Loops over past shots and
give total history using
shot.history

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The Basketball Shot Predictor allows users to simulate basketball shot trajectories by adjusting physics parameters like launch angle, ball speed, initial height, and distance. The app calculates the flight path, determining whether the shot hits or misses the hoop.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	**Shot** **current_angle** (float): Stores the currently selected launch angle in degrees. Modified at the start by user **current_speed** (float): Stores the launch velocity of the ball in meters per second. Modified when the user adjusts the speed at the beginning. **Current_height** (float): Stores the height of the shooter. It will stay the same and only be altered by the user at the start of each run. **shot_history** list[ShotResult]: A list of the second dataclass. Modified by appending a new record every time the user clicks "Fire Shot".
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	**ShotResult** **outcome** (str): Summary outcome of the shot: "Swish!", "Short", "Long", "Blocked" **peak_height** (float): Maximum height

	Distance_from (float): measures the final distance from the hoop. If it goes in, the distance is None. Changed after each shot.
Routes (minimum 7) The names of your route functions	index: Main home route that initializes default state update_angle: Handles user input change for launch angle. update_speed: Handles user input change for ball speed. update_distance: Handles user input change for distance from the hoop. toggle_defender: Toggles defender presence on or off. fire_shot: Triggers trajectory calculations, updates outcome state, and appends to history. Reset: Resets the shot list back to empty.
Pages (minimum 5) A brief description of the pages in your site	**Simulator Page**: Main interface containing input controls **Shot History Page**: Displays a table of all past attempted shots with their outcomes. **Graph Page.** This page shows the graph of the latest shot and how the trajectory would have looked using NumPy **Final Page**: Provides the output of the current shot. **Stats & Analytics Page**: Calculates summary statistics based on the past shots list (shooting percentage, average peak height).
User Input (minimum 3)	**TextBox**: Captures the value for ball speed to calculate forward and vertical velocity.

Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	**SelectBox**: Allows selecting predefined shot distances or to input your own. **CheckBox**: Toggles whether a defender is present on the court
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	**Defender Clearance Check**: checks if ball height is greater than the defender height. If true, the outcome will be a miss and blocked **Distance_long check:** checks to see if the ball height and speed are greater than the amount needed to surpass or make it to the hoop **Distance_short check:** checks to see if the ball height and speed are shorter than the amount needed to make it to the hoop **Angle check**: checks to see if the angle is great enough for the ball to be above the hoop.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	**Iteration Target**: **Iterates through state.shot_history (list of ShotResult instances).** **Description & Purpose**: The loop goes through each past ShotResult object to construct the historical stats table and calculate overall shooting percentage. It is meaningful because it shows data about the past shots and gives data to the user.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I used an LLM for this ending part of the table assignment, for the loops part. It allowed me ot come up with a meaningful loop that I can use in my project. I also used an LLM to help me come up with meaningful names and inputs.

I want to make a basketball shot predictor using physics. What features should I include

AI: *You could let users adjust launch angle, speed, starting height, and distance, then graph the trajectory. To make it fun, you could add an optional defender obstacle to see if the shot gets blocked.*

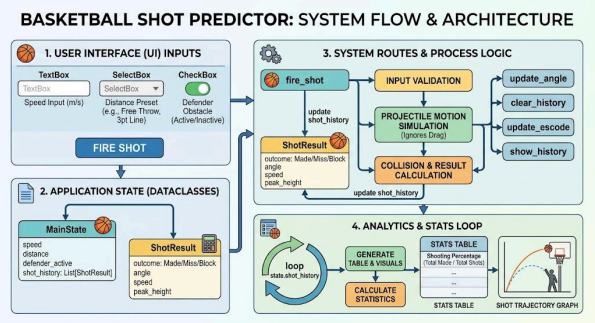
Can I simplify it by ignoring air resistance and focusing on basic speed, angle, and distance?

> ***AI:*** *Absolutely. Standard projectile motion without drag keeps the physics straightforward while still giving realistic arc paths.*

I need to fill out my project proposal table. Can you help me brainstorm names for my inputs, routes, dataclasses, and a loop that iterates over saved shot history?

> ***AI:***
>
> - ***Inputs:*** *Speed (`TextBox`), Preset Distance (`SelectBox`), Defender Toggle (`CheckBox`).*
> - ***Dataclasses:*** *Main `State` (stores inputs and `shot_history`) and `ShotResult` (stores outcome, angle, speed, peak height).*
> - ***Routes:*** `index`, `update_angle`, `update_speed`, `update_distance`, `toggle_defender`, `fire_shot`, `clear_history`.
> - ***Loop:*** *Loop over `state.shot_history` (a list of `ShotResult` instances) to calculate shooting percentage and generate the stats table on the analytics page.*

Also help me make a chart that will give a brief understanding of the program.



(include a flow chart like this)

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas